

ArchIntent



Project Proposal

Submitted by

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Brief Details About Project

Project Name	ArchIntent
Category Name (If industrial based project)	
Client Details (If industrial based project)	Name:
	Contact No:
Date of Commencement	December 1, 2025
Expected Date of Completion	June 30, 2026
Group members name with CGPA (till 6th semester and registration number)	Name Registration C.G.P.A
	Muhammad
	Dawood Khan FA22-BSE-067 3.3
	Huzaifa Imran FA22-BSE-075 2.5
Average CGPA	2.9
Implementation Tool	ReactJS, Laravel, MySQL, WhisperASR, SpaCy, S-BERT, Visual Studio Code
Graphics Development Tools	Figma
Documentation Tools	MS Word, LaTeX
Name of Platform for Implementation	Vercel and Render
Proposed Supervisor Name (From SE)	Dr. Shamila
Proposed Co-Supervisor Name (From SE)	Engr. Asim Javed

1 Project Summary/Introduction

The architecture and construction industry, a cornerstone of urban development, is currently hindered by significant fragmentation and inefficiency. A persistent disconnect exists among key stakeholders: clients with a vision, architects with the design expertise, and construction companies with the execution capabilities. This "discovery gap" forces clients to navigate a complex, manual landscape where articulating their needs in technical terms is difficult, and finding reliable professionals is often a gamble based on unverified referrals. Simultaneously, talented architects and contractors struggle to showcase their specialized portfolios to the right audience, often relying on inefficient general marketplaces that fail to capture the nuance of design style or the security required for high-stakes construction agreements.

ArchIntent addresses these systemic failures by introducing an AI-augmented, three-sided marketplace that unifies the pre-construction pipeline. By leveraging Natural Language Processing (NLP) to decode client intent and semantic matching to align design sensibilities, the platform replaces manual searching with intelligent recommendation. It fosters a secure ecosystem where trust is established through verified portfolios and digital contract formalization, ensuring that every connection is based on genuine compatibility and transparency. This solution not only simplifies the client's journey from concept to contract but also empowers professionals with a stream of high-quality leads, effectively modernizing the way architectural projects are initiated and collaborative teams are formed.

1.1 Problem Statement

The architecture and construction industry lacks a unified digital ecosystem, making it difficult for clients to find professional architects and verified contractors. Existing platforms fail to bridge the pre-contract gap, leaving processes vulnerable to fraud. There is a need for an intelligent solution that enables seamless collaboration, secure agreements, and intent-based design discovery.

1.2 Proposed Solution

Our proposed solution is an online platform that bridges architects, construction companies, and the general public. The platform allows architects to create professional portfolios showcasing their designs, while users can browse, purchase, or directly contact them for customized projects. Construction companies can register to collaborate with users and architects, streamlining the project realization process.

The platform will also feature intelligent components such as semantic search and keyword extraction to help users find relevant designs easily.

1.3 Motivation & Societal Impact

The primary motivation for the **ArchIntent** platform is to address the pre-contract vacuum in the AEC sector. Traditional methods fail due to the NLP Gap (translating non-technical client needs) and the Contract Gap (manual, opaque agreement procedures) [2]. Furthermore, the rise of digital multi-sided platforms has revolutionized other sectors, yet the construction industry lacks a unified ecosystem for seamless stakeholder collaboration [8]. Recent studies in automated cost estimation further highlight the need for integrated financial transparency early in the design phase [5]. The subsequent review focuses on existing solutions to pinpoint the precise competitive advantage of **ArchIntent**.

2 Competitive Analysis & Market Gap

Category	Incumbent Example	Limitation / Core Gap	ArchIntent Advantage
Primary Function	Procore (Post-Contract)	Designed for execution (B2B); zero client acquisition services.	Pre-Contract Marketplace: Unifies Client, Architect, & Contractor from concept phase [7].
Intent Decoding	Togal.AI (Technical)	Processes CAD inputs only; cannot decode layman natural language (NLP Gap) [6].	Decoded Intent: Converts vague voice/text into structured parameters via SpaCy/Whisper [2, 3].
Style Match Quality	Upwork (Keyword Search)	Limited to basic keywords; fails to match semantic design ethos.	Semantic Matching: Uses S-BERT to vectorize client needs against portfolios [4, 10].
Contracting & Trust	Generic Directories	Manual negotiation leads to opacity and fraud risks (Contract Gap).	Digital Agreements (SDG 16): Secure, auditable workflow for signing project agreements [3, 4].

Table 1: **Competitive Feature Comparison: ArchIntent vs. Market Leaders**

This comparative analysis demonstrates that existing platforms in the AEC sector address isolated pain points but fail to deliver an integrated pre-contract solution. ArchIntent fills this critical market gap by combining intelligent intent

recognition, semantic matching, and streamlined digital contracting in a single unified platform [3].

3 Proposed Goals/Objectives

3.1 Main Goal

To develop an intelligent web platform that connects architects, construction companies, and the public for efficient design discovery, communication, and project collaboration.

3.2 Specific Objectives

- Allow architects to create and manage design portfolios with semantic tagging.
- Enable users to browse, filter, and buy designs or contact architects directly through natural language queries.
- Provide construction companies a channel to connect with potential clients through verified profiles.
- Integrate AI-based search and voice query features for easy access using WhisperASR and S-BERT.
- Ensure secure authentication, data management, and smooth communication between all stakeholders.
- A real-time communication hub enabling secure interaction between clients, architects, and contractors.

4 Implementation Method

4.1 Development Methodology

We will use the **Incremental Model**, where the system is developed and delivered in multiple, functional increments. Each increment adds more features and improves upon the previous version. This approach allows early testing, easier debugging, and faster delivery of working modules.

Initially, a core functional system (such as registration, login, and basic design browsing) will be implemented. In subsequent increments, additional features like AI-based search, voice queries, and real-time communication will be integrated [1].

4.2 Major Modules

- **User Module:** Handles user registration, authentication, browsing of architectural designs, and initiating contact with architects.
- **Architect Module:** Allows design uploads, portfolio management, and direct communication with clients using semantic tagging.
- **Construction Module:** Enables construction companies to list services and connect with clients through verified credentials.
- **Admin Module:** Provides administrative control for user verification, data moderation, and analytics.
- **AI Module:** Integrates WhisperASR for speech-to-text, SpaCy for keyword extraction, and S-BERT for semantic search, enabling intelligent user interactions.

4.3 Incremental Development Phases

1. **Increment 1:** User registration, login, and portfolio viewing.
2. **Increment 2:** Design upload and management by architects with semantic tagging.
3. **Increment 3:** Construction company integration and collaboration features.
4. **Increment 4:** AI-based semantic and voice search functionalities.
5. **Increment 5:** Final testing, optimization, and deployment.

4.4 Data Flow

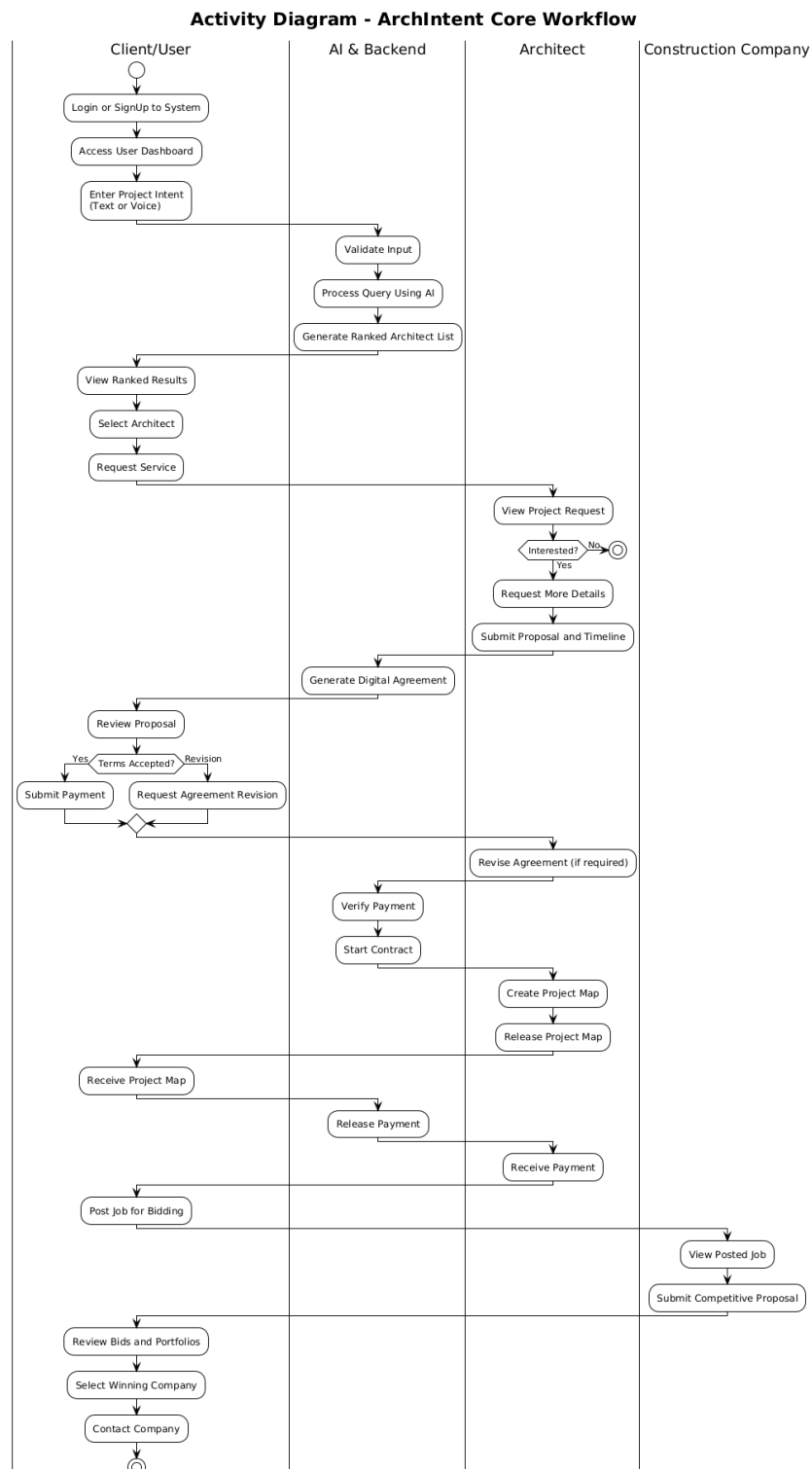


Figure 1: System Activity Diagram

5 Benefits

- Simplifies discovery of architects and construction services through intelligent matching [10].
- Provides digital portfolios for professional visibility and trust-building [3].
- Enables secure, efficient communication and collaboration through digital contracting [2].
- Reduces time and effort in design selection and contractor hiring via semantic search [4].
- Encourages local innovation and economic activity in construction [8].
- Bridges the NLP Gap by converting natural language requirements into structured parameters [6].

6 Technical Details of Final Deliverable

6.1 Final Product Includes

- Responsive web application (desktop and mobile view).
- Interactive dashboards for users, architects, and construction companies.
- Real-time chat feature with secure messaging.
- AI-based search and voice query system using WhisperASR and S-BERT.
- Admin control panel for user and data management with verification workflows [7].

6.2 Platforms Supported

Web (Cross-Platform Browser-Based)

6.3 APIs Integrated

- **Whisper API** - Speech-to-Text for natural language and voice-based input
- **Stripe API** - Secure online payment processing, billing, and transaction management

7 Target Industries

- I. Architecture** This industry is central to the platform, as **ArchIntent** provides architects with a dedicated space to host professional portfolios and showcase designs. It resolves the “client acquisition” struggle by allowing architects to be easily discovered by users through semantic search and design tags.
- II. Construction** The platform targets this industry by offering a verified marketplace where construction companies can list their services and connect directly with potential clients. It streamlines the project realization process by allowing contractors to join the workflow early, fostering collaboration between builders, architects, and property owners.
- III. Real Estate Tech (PropTech)** **ArchIntent** innovates within this sector by digitizing the “pre-contract vacuum”—the phase before formal agreements are signed—which is currently manual and opaque. It introduces digital contracting and efficient matching tools to modernize how real estate design and construction services are procured.

8 Tools and Technologies

Category	Tools/Technologies
Frontend	ReactJS, Tailwind CSS
Backend	Laravel (PHP Framework)
Database	MySQL
AI & NLP	WhisperASR, SpaCy, Sentence Transformers (S-BERT)
Authentication	Laravel Sanctum
Payment Gateway	Stripe
Version Control	GitHub
Development	VS Code
Hosting	Render
Communication	Laravel Echo

Table 2: Technology Stack

9 Sustainable Development Goals (SDGs)

SDG Goal	Mapping to Requirements
1. SDG 9 – Industry, Innovation & Infrastructure	Promotes digital innovation in architecture and construction via AI matching.
2. SDG 8 – Decent Work & Economic Growth	Creates employment and collaboration opportunities in the AEC sector.
3. SDG 11 – Sustainable Cities & Communities	Encourages sustainable planning through verified professional networks.
4. SDG 16 – Peace, Justice & Strong Institutions	Provides transparent, auditable digital contracting to reduce fraud.

Table 3: Sustainable Development Goals Alignment

10 Milestones

Month	Deliverables
Month 1	Requirement gathering, feasibility study, and UI/UX design
Month 2	Increment 1: Core features – authentication and profile setup
Month 3	Increment 2: Architect portfolio and design upload module with semantic tagging
Month 4	Increment 3: Construction company module, Real Time communicatio and admin controls with verification
Month 5	Increment 4: AI module integration (WhisperASR, SpaCy, S-BERT) and Payment Integration
Month 6	Increment 5: Final testing, optimization, deployment, and report writing

Table 4: Project Milestones Timeline

11 Equipment Details

Item	Quantity	Cost (PKR)	Purpose
Laptop (Existing)	3	–	Development and testing
Internet Connection	–	3,000/month	Online tools and collaboration
Hosting & Domain	1	5,000	Deployment
WhisperASR API	–	Free (Open Source)	Voice processing
SpaCy / S-BERT	–	Free	NLP and semantic search
Stripe API	–	Free (Pay-as-you-go)	Secure payment processing

Table 5: Equipment and Resource Requirements

12 Project Schedule Plan

Activity	Duration
Requirement Analysis	3 weeks
System Design	3 weeks
Core Implementation (Frontend + Backend)	6 weeks
Payment Gateway Integration (Stripe)	2 weeks
AI Integration	4 weeks
Testing & Debugging	3 weeks
Report Writing & Presentation	3 weeks

Table 6: Detailed Project Schedule

13 Complex Engineering Problem Attributes

This project qualifies as a complex engineering problem because it involves:

- Multi-disciplinary integration of software engineering, AI, NLP, and database systems.
- Complex data management and real-time interaction among multiple stakeholders.
- AI-based intelligent features requiring deep algorithmic processing with S-BERT and SpaCy.
- Scalability and performance optimization for large datasets and user bases.
- Security and authentication challenges across multiple user roles with digital contracting.
- Natural language processing challenges in bridging the NLP Gap between technical and non-technical users.

13.1 Complexity Justification

Aspect	Complexity
Technical Depth	Integration of machine learning models (S-BERT), NLP (SpaCy), and speech recognition (WhisperASR) for semantic understanding
Scale	Handling multiple user types with different permissions and workflows simultaneously in a three-sided marketplace
Interdependency	Real-time communication between frontend, backend, AI services, and database with secure contracting workflows
Innovation	Novel application of semantic search in architecture domain bridging the NLP Gap
Security	Multi-role authentication, data privacy, and secure transactions with auditable digital agreements

Table 7: Engineering Complexity Analysis

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